

Assignment /

Assignment #: 14

Topic : Collections

1. Write a program that asks how many names the user wants to input, asks for those names, and shows the names afterward. But it should only show names that are longer than 5 characters using Array List

Sample Input: How many names do you want to input? 4
Enter name #1: Anu
Enter name #2: Pragathi
Enter name #3: Charlie
Enter name #4: Shiny

Sample Output: Names longer than 5 characters:
Pragathi
Charlie

2. Suppose you have a set s containing some strings. Write a code fragment that uses an iterator to print out each element in s and also compute the result of concatenating all of these strings together.

Sample Input & Output: Elements in the Set : Java
Elements in the Set : Hello
Elements in the Set : Programming
Elements in the Set : World
Concatenated result: Java Hello Programming World

3. Write a code and print the count (number of occurrences) of characters a,c,o, and s in the string (using a hash Map).
 - a. String text = "Write Java code to define yet another Set s. Insert 3 floating point numbers in s, and using an iterator, find the sum of the numbers in s."

Sample Input & Output: Character counts:
a: 7
c: 1
s: 8
o: 7

4. Julie has created a HashMap containing the name and marks as key-value pairs of 6 students. Write a program, with separate user-defined functions to perform the following operations:
- Push the keys (name of the student) of the hash map into a stack, where the corresponding value(marks) is greater than 75.
 - Pop and display the content of the stack.

Sample Input & Output:

Student Marks in the HashMap: {Mano=85, Shan=85, John=55, Anu=60, Aju=90, Frank=80}

Student Marks above 75 in the stack:

Frank
Aju
Shan
Mano

5. Write a program that reads a sequence of numbers and displays all the numbers in sorted order without duplicates. then: – Prints the average of the numbers. – Prints the highest and lowest number. – Filters out all of the even numbers (ones divisible by 2).

Sample Input:

Enter a number (or type 'done' to finish): 4
Enter a number (or type 'done' to finish): 7
Enter a number (or type 'done' to finish): 7
Enter a number (or type 'done' to finish): 5
Enter a number (or type 'done' to finish): 3
Enter a number (or type 'done' to finish): 9
Enter a number (or type 'done' to finish): 8
Enter a number (or type 'done' to finish): done

Sample Output:

Sorted unique numbers: [3, 4, 5, 7, 8, 9]
Average: 6.0
Lowest: 3
Highest: 9
Odd numbers: [3, 5, 7, 9]

6. Create a class, Person. A person has a first name and a last name. Write a main program that creates an ArrayList with a number of people (e.g., Priscilla Wagner, Tom Parker, Elvis Presley). Sort the ArrayList based on the first name and show the result: Elvis Presley Priscilla Wagner Tom Parker. Use a Comparator or Comparable Interface.

Sample Input & Output: Sorted people by first name:
Elvis Presley
Priscilla Wagner
Tom Parker